

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 7: Ethics and Ownership

Subtopic 7.1: Ethics and Ownership

Concept 7.1.3: Software Licensing

Total Questions: 14

Mark Schemes begin on Page 7

Question 1 | Source: 9618_w25_qp_12 (Q2.a)

2 (a) Complete the table by describing each term.

Term	Description
Copyright	
Open Source (Initiative)	
Shareware	
Software Licence	

[4]

Question 2 | Source: 9618_w25_qp_12 (Q10.b)

10 A programmer is developing a computer program.

(b) The programmer releases the program as Free Software.

Describe what is meant by Free Software.

[2]

Question 3 | Source: 9618_w25_qp_12 (Q10.c)

10 A programmer is developing a computer program.

- (c) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

..... [1]

Question 5 | Source: 9618_s25_qp_12 (Q3.c.i)

- 3 A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

- (c) The computer program is released under a commercial software licence.

- (i) Describe the features of a commercial software licence.

.....
.....
.....
.....
.....
..... [3]

Question 6 | Source: 9618_s25_qp_12 (Q3.c.ii)

- 3 A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

- (c) The computer program is released under a commercial software licence.

- (ii) Explain the reasons why an open source software licence might **not** be appropriate for the computer program described.

.....
.....
.....
.....
.....
..... [3]

Question 7 | Source: 9618_w24_qp_11 (Q7.a)

7 Software is distributed with a licence.

(a) Give **two** benefits of distributing software using a shareware software licence.

1

.....

2

.....

[2]

Question 8 | Source: 9618_w24_qp_11 (Q7.b)

7 Software is distributed with a licence.

(b) Give **two** benefits of distributing software using a commercial software licence.

1

.....

2

.....

[2]

Question 9 | Source: 9618_w24_qp_12 (Q5.b.i)

5 A computer programmer is learning about the importance of acting ethically at work.

(b) The programmer writes a program to be used in a business. The source code of the program can be modified by the business. The business pays the programmer for the maintenance and security updates.

(i) Identify **one** suitable type of software licence for the program **and** describe the reasons for your choice.

Type of software licence

Reasons for choice

.....

.....

.....

[3]

Question 10 | Source: 9618_w24_qp_13 (Q5.a)

5 A computer programmer writes programs that are distributed with a software licence.

- (a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	
Users can try the software before buying it. Users may not be able to access all features during the trial.	
Software is usually free of charge. Users can access the source code and alter the program to their needs.	

[3]

Question 11 | Source: 9618_w23_qp_13 (Q6.a)

- 6 (a) State **two** benefits to a programmer of distributing a program using a shareware licence.

1

.....

2

.....

[2]

Question 12 | Source: 9618_s23_qp_13 (Q5.b)

- 5 A programmer is developing a computer game in a high-level language to sell to the public.

- (b) The programmer needs to publish the game under a software licence so that it can be sold to the public.

Identify the **most appropriate** type of software licence for the game **and** justify your choice.

Licence

Justification

.....

.....

.....

.....

.....

[4]

Question 13 | Source: 9618_w22_qp_13 (Q8.a.i)

8 (a) (i) Explain why some programs are distributed under an open source licence.

.....

.....

.....

..... [2]

Question 14 | Source: 9618_w22_qp_13 (Q8.a.ii)

(ii) Explain how a programmer benefits from distributing a program under a commercial licence.

.....

.....

.....

..... [2]

Question 15 | Source: 9618_s21_qp_12 (Q2.c)

2 Aisha manages a team of software developers.

(c) The final game will be released under a licence.

Tick (✓) **one or more** boxes in each row to identify the licence(s) each statement describes.

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code				
The user must always pay before being able to use the software				
The user can redistribute the software				
The user always gets a trial period				

[4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q2.a)

2(a)	1 mark for each correct row, max 4 marks	4										
<table><tr><th>Term</th><th>Description</th></tr><tr><td>Copyright</td><td>Gives the holder the legal right of ownership of intellectual property</td></tr><tr><td>Open Source (Initiative)</td><td>Software can be freely copied, distributed or adapted</td></tr><tr><td>Shareware</td><td>Gives the user a trial period / limited functionality before a payment/sign up may be required</td></tr><tr><td>Software Licence</td><td>A (legal) agreement that defines the rights and terms of usage for the software</td></tr></table>			Term	Description	Copyright	Gives the holder the legal right of ownership of intellectual property	Open Source (Initiative)	Software can be freely copied, distributed or adapted	Shareware	Gives the user a trial period / limited functionality before a payment/sign up may be required	Software Licence	A (legal) agreement that defines the rights and terms of usage for the software
Term	Description											
Copyright	Gives the holder the legal right of ownership of intellectual property											
Open Source (Initiative)	Software can be freely copied, distributed or adapted											
Shareware	Gives the user a trial period / limited functionality before a payment/sign up may be required											
Software Licence	A (legal) agreement that defines the rights and terms of usage for the software											

Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q10.b)

10(b)	1 mark per bullet point, max 2 - The software is free of restrictions - The software is not necessarily free of charge - The <u>source code</u> comes with the program // A user can edit the <u>source code</u> to add functionality / suit their needs - Users can share the software with others	2
-------	--	----------

Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q10.c)

10(c)	1 mark for a correct answer Digital Signature	1
-------	--	----------

Mark Scheme for Question 4 | Source: 9618_w25_ms_13 (Q8.c)

8(c)	1 mark per bullet point, max 2 marks - Open Source (Initiative) - Free Software (Foundation)	2
------	---	----------

Mark Scheme for Question 5 | Source: 9618_s25_ms_12 (Q3.c.i)

3(c)(i)	1 mark each to max 3 • The program is usually sold for a fee • Users have no access to source code • It protects the intellectual rights of the programmer / developer • Users cannot legally alter or share the program • There are limitations on the use of the software e.g. number of times it can be installed • Support and maintenance is usually provided e.g. updates, bug fixes etc.	3
---------	--	----------

Mark Scheme for Question 6 | Source: 9618_s25_ms_12 (Q3.c.ii)

3(c)(ii)	1 mark each to max 3 e.g. • The source code would be accessible therefore it could be changed • which might lead to the program outputting an incorrect response • so people could purchase the wrong item • which might have serious consequences // by example e.g. allergies	3
----------	---	----------

Mark Scheme for Question 7 | Source: 9618_w24_ms_11 (Q7.a)

7(a)	1 mark for each bullet point (max 2) • The user can test/trial the program before purchasing to see whether it meets needs • There is no initial cost • Users who try the software may spread the word so the developer gets more custom • The developer can get feedback to improve the program	2
------	---	----------

Mark Scheme for Question 8 | Source: 9618_w24_ms_11 (Q7.b)

7(b)	1 mark for each bullet point (max 2) • The owner/developer can earn a fee • Comprehensive support is usually offered so problems can be easily solved • The software is usually well tested so few bugs	2
------	--	----------

Mark Scheme for Question 9 | Source: 9618_w24_ms_12 (Q5.b.i)

5(b)(i)	Type of software licence 1 mark for - Open Source Initiative // Free Software Foundation Reason for choice: 1 mark for each bullet point (max 2) - Allows program source code to be examined ... so that the business can adapt/modify the program to meet their needs - Allows the programmer to earn money for the software and updates	3
---------	--	----------

Mark Scheme for Question 10 | Source: 9618_w24_ms_13 (Q5.a)

5(a)	1 mark for each correct licence		3
	Statement	Software Licence	
	A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	Commercial	
	Users can try the software before buying it. Users may not be able to access all features during the trial.	Shareware	
	Software is usually free of charge. Users can access the source code and alter the program to their needs.	Open Source // Free software	

Mark Scheme for Question 11 | Source: 9618_w23_ms_13 (Q6.a)

6(a)	1 mark for each bullet point (max 2). - Users are able to trial the program and may offer the programmer constructive feedback More people might try the program because the trial is free of charge - Allows bugs to be found and corrected on a wide range of computer system configurations improving the programmer's original program - Users that find the trial useful will buy the program so programmer gets income - Allows the program to be copyrighted and so protects the programmer's intellectual property rights // no illegal modification allowed	2
------	--	----------

Mark Scheme for Question 12 | Source: 9618_s23_ms_13 (Q5.b)

5(b)	1 mark for appropriate licence; 1 mark for each point to max 3 • Commercial software licence • User has to pay for the product so the programmer can gain an income • Enables the program to be copyrighted • ... so the user cannot legally edit the program // the programmer retains control over product • ... and can take legal action against people who attempt to illegally copy it /sell it on • Shareware licence • Enables the program to be copyrighted • The user cannot legally edit the program so the developer retains control over product • User can try the program for free and then pay for the full game which allows the programmer to gain an income • so more people can experience it and therefore be more likely to buy it	4
------	--	----------

Mark Scheme for Question 13 | Source: 9618_w22_ms_13 (Q8.a.i)

8(a)(i)	1 mark for each bullet point (max 2): • to allow users to customise the code • to allow errors to be reported / identified / fixed by users • to allow additional features to be added to the code • to allow for collaboration	2
---------	---	----------

Mark Scheme for Question 14 | Source: 9618_w22_ms_13 (Q8.a.ii)

8(a)(ii)	1 mark for each correct point (max 2) Example: • enables the program to be copyrighted • prevents illegal changes to the program / protects the source code • prevents illegal copies of the program being made • a fee can be charged for the program	2
----------	---	----------

Mark Scheme for Question 15 | Source: 9618_s21_ms_12 (Q2.c)

2(c)

1 mark for each correct column

4

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code	✓	✓		
The user must always pay before being able to use the software				✓
The user can redistribute the software	✓	✓	✓	
The user always gets a trial period			✓	